

Flu Project Extension

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This project is an extension of the national influenza forecasting model that explores a hierarchical structure across the ten United States Department of Health and Human Services (HHS) regions. The motivating idea behind this is that flu does not behave identically everywhere. Some regions run hotter and some colder, and the strength of week-to-week momentum may itself differ between regions. A hierarchical model captures this by allowing each region its own parameters while sharing information across regions using partial pooling, producing more stable estimates than fitting ten separate models would. The central question focused in this project is as follows: does modeling regional structure meaningfully improve forecasting, and how do the regions actually differ?

Data and Feature Construction

```
#Libraries
library(tidyverse)
library(brms)
library(ggplot2)
```

```
#Import dataset into R
pa = "H:/RStudio Files/data/FluViewPhase2DataRegional/ILINetRegional.csv"
df = read.csv(pa, skip = 1)
colnames(df)
```

```
## [1] "REGION.TYPE"      "REGION"           "YEAR"
## [4] "WEEK"             "X..WEIGHTED.ILI"  "X.UNWEIGHTED.ILI"
## [7] "AGE.0.4"          "AGE.25.49"        "AGE.25.64"
## [10] "AGE.5.24"         "AGE.50.64"        "AGE.65"
## [13] "ILITOTAL"         "NUM..OF.PROVIDERS" "TOTAL.PATIENTS"
```

Similar to the national influenza project, this data is the regional ILINet export from CDC, covering all ten HHS regions weekly from 1997 to 2026. Cleaning is very similar to that of the national project: convert “X”-coded missing values to NA, rename columns, and use coalesce to reconcile the age bands. One addition to the structure is converting region to a factor, which is required by the model to treat it as a grouping variable.

```
num_cols = c("X..WEIGHTED.ILI", "X.UNWEIGHTED.ILI", "AGE.0.4", "AGE.5.24",
             "AGE.25.49", "AGE.25.64", "AGE.50.64", "AGE.65", "ILITOTAL",
             "NUM..OF.PROVIDERS", "TOTAL.PATIENTS")

df_clean = df %>%
  mutate(across(all_of(num_cols), ~ suppressWarnings(as.numeric(.)))) %>%
  rename(
    region_type = REGION.TYPE,
    region = REGION,
    year = YEAR,
    week = WEEK,
    wili = X..WEIGHTED.ILI,
    ili = X.UNWEIGHTED.ILI,
    age_0_4 = AGE.0.4,
    age_5_24 = AGE.5.24,
    age_25_49 = AGE.25.49,
    age_25_64 = AGE.25.64,
    age_50_64 = AGE.50.64,
    age_65 = AGE.65,
    ili_total = ILITOTAL,
    providers = NUM..OF.PROVIDERS,
    total_pat = TOTAL.PATIENTS
  )

df_clean = df_clean %>% mutate(region = factor(region))

summary(df_clean)
```

```
##      region_type      region      year      week
## Length :14980 Region 1 :1498 Min. :1997 Min. : 1.00
## N.unique : 1 Region 10:1498 1st Qu.:2004 1st Qu.:13.00
## N.blank : 0 Region 2 :1498 Median :2012 Median :26.00
## Min.nchar: 11 Region 3 :1498 Mean :2012 Mean :26.55
## Max.nchar: 11 Region 4 :1498 3rd Qu.:2019 3rd Qu.:40.00
## Region 5 :1498 Max. :2026 Max. :53.00
## (Other) :5992
##      wili      ili      age_0_4      age_25_49
## Min. : 0.0000 Min. : 0.0000 Min. : 0.0 Min. : 0.00
## 1st Qu.: 0.6933 1st Qu.: 0.7608 1st Qu.: 35.0 1st Qu.: 84.75
## Median : 1.3144 Median : 1.3463 Median : 175.0 Median : 273.50
## Mean : 1.8031 Mean : 1.7847 Mean : 472.6 Mean : 718.17
## 3rd Qu.: 2.3113 3rd Qu.: 2.2787 3rd Qu.: 549.2 3rd Qu.: 693.00
## Max. :15.9426 Max. :14.2383 Max. :12115.0 Max. :16107.00
## NAs :5320
##      age_25_64      age_5_24      age_50_64      age_65
## Min. : 0.0 Min. : 0.0 Min. : 0.0 Min. : 0.0
## 1st Qu.: 5.0 1st Qu.: 52.0 1st Qu.: 34.0 1st Qu.: 8.0
## Median : 38.0 Median : 237.0 Median : 114.0 Median : 42.0
## Mean : 108.5 Mean : 700.1 Mean : 289.5 Mean : 171.5
## 3rd Qu.: 124.0 3rd Qu.: 699.0 3rd Qu.: 283.0 3rd Qu.: 140.0
## Max. :3245.0 Max. :25205.0 Max. :6969.0 Max. :7847.0
## NAs :8710 NAs :5320
##      ili_total      providers      total_pat
## Min. : 0 Min. : 0.0 Min. : 0
## 1st Qu.: 143 1st Qu.: 55.0 1st Qu.: 13922
## Median : 682 Median :144.0 Median : 52880
## Mean : 2039 Mean :177.1 Mean : 84708
## 3rd Qu.: 2025 3rd Qu.:239.0 3rd Qu.:107559
## Max. :58023 Max. :976.0 Max. :745687
##
```

The data comprise 14,980 weekly observations, ten regions of 1,498 weeks each, with weighted ILI fully populated at the regional level. A feature that is important to note is that regional wILI is even more right-skewed than at the national level (maximum of 16 versus 8), since individual regions have local peaks that are sharper than the national average tends to smooth out. This further reinforces choosing a skewed and strictly positive likelihood for the model.

```
model_df = df_clean %>%
  arrange(region, year, week) %>%
  group_by(region) %>%
  mutate(
    sinYear = sin(2 * pi * week / 52),
    cosYear = cos(2 * pi * week / 52),

    wili_lag1 = lag(wili, 1),
    wili_lag2 = lag(wili, 2),

    peds = age_0_4 + age_5_24,
    adults = coalesce(age_25_49 + age_50_64, age_25_64),
    elderly = age_65,
    age_denom = peds + adults + elderly,
    share_peds_lag1 = lag(peds / age_denom, 1),
```

```

share_elderly_lag1 = lag(elderly / age_denom, 1),

pandemic_era = ifelse(year %in% c(2020, 2021), 1, 0),

wili_next = lead(wili, 1)
) %>% ungroup()

```

Creating new features follows the same approach from the national project with one crucial difference: all lags and leads must be computed within each of the ten regions separately. Grouping by region before lagging prevents one region's final week from being treated as the prior week of the next region, which would corrupt the features without warning.

```

model_df %>%
  group_by(region) %>%
  slice(1) %>%
  select(region, year, week, wili, wili_lag1, wili_lag2) %>%
  print(n = 10)

```

```

## # A tibble: 10 x 6
## # Groups:   region [10]
##   region    year week  wili wili_lag1 wili_lag2
##   <fct>    <int> <int> <dbl>      <dbl>      <dbl>
## 1 Region 1  1997    40 0.499         NA         NA
## 2 Region 10 1997    40 4.74         NA         NA
## 3 Region 2   1997    40 0.375         NA         NA
## 4 Region 3   1997    40 1.35         NA         NA
## 5 Region 4   1997    40 0.400         NA         NA
## 6 Region 5   1997    40 1.23         NA         NA
## 7 Region 6   1997    40 1.02         NA         NA
## 8 Region 7   1997    40 0.872         NA         NA
## 9 Region 8   1997    40 0.516         NA         NA
## 10 Region 9  1997    40 1.81         NA         NA

```

These results confirm grouping worked. The first observation of every region shows missing lags, since no prior week exists within each region. Had the grouping failed, only the very first row of the entire dataset would have been missing.

Using the same logic as in the national analysis, the train/test split is made at the 2022 boundary to prevent look-ahead leakage. Both the training and testing sets retain all ten regions, confirming that no region was lost to missingness during filtering.

```

threshold = 3.0

feat_base = c("wili_lag1", "wili_lag2", "sinYear", "cosYear")
feat_age = c(feat_base, "share_peds_lag1", "share_elderly_lag1")

cutoff = 2022
train_df = model_df %>%
  filter(year < cutoff) %>%
  drop_na(all_of(c("wili_next", feat_age)))
test_df = model_df %>%
  filter(year >= cutoff) %>%

```

```
drop_na(all_of(c("wili_next", feat_age)))

nrow(train_df); nrow(test_df)

## [1] 11638

## [1] 2310

n_distinct(train_df$region); n_distinct(test_df$region)

## [1] 10

## [1] 10

train_df = train_df %>% mutate(wili_next_pos = wili_next + 0.01)
test_df = test_df %>% mutate(wili_next_pos = wili_next + 0.01)
```

The Hierarchical Model

The hierarchical model adds a region-level term, $(1 + \text{wili_lag1} \mid \text{region})$, to the Gamma regression. This allows each region its own intercept (its baseline flu level) and its own slope on the previous week's activity (its momentum), while estimating how much these vary by region. That estimated variation is what drives partial pooling: regions with abundant, clean data retain estimates close to their own pattern, while noisier regions are pulled toward the overall average, borrowing strength from others. The intercept and the strongest predictor are the only coefficients allowed to vary by region, which is a deliberate choice to keep the model tractable rather than letting all of the coefficients vary at the same time.

```
fit_hier = brm(
  bf(wili_next_pos ~ wili_lag1 + wili_lag2 + sinYear + cosYear + share_peds_lag1
    + share_elderly_lag1 + (1 + wili_lag1 | region)),
  data = train_df,
  family = Gamma(link = "log"),
  chains = 4, cores = 4, iter = 2000, seed = 123,
  refresh = 0
)
```

```
summary(fit_hier)
```

```
## Family: gamma
## Links: mu = log
## Formula: wili_next_pos ~ wili_lag1 + wili_lag2 + sinYear + cosYear + share_peds_lag1 + share_elderly_lag1
## Data: train_df (Number of observations: 11638)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~region (Number of levels: 10)
##
```

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	
sd(Intercept)	0.37	0.10	0.23	0.61	1.01	881	
sd(wili_lag1)	0.10	0.03	0.06	0.17	1.01	913	
cor(Intercept,wili_lag1)	-0.98	0.02	-1.00	-0.92	1.00	1045	

```
##
```

	Tail_ESS
sd(Intercept)	1536
sd(wili_lag1)	1581
cor(Intercept,wili_lag1)	1748

```
##
```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.25	0.13	-0.50	0.00	1.00	889	1427
wili_lag1	0.35	0.04	0.29	0.42	1.00	886	1521
wili_lag2	-0.02	0.01	-0.04	-0.01	1.00	5170	3017
sinYear	-0.02	0.01	-0.03	-0.00	1.00	5093	3012
cosYear	0.40	0.01	0.38	0.41	1.01	5351	2978
share_peds_lag1	0.05	0.05	-0.05	0.15	1.00	3812	2745
share_elderly_lag1	-0.57	0.12	-0.81	-0.33	1.00	3694	2795

```
##
```

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
shape	4.18	0.05	4.08	4.28	1.00	4503	2757

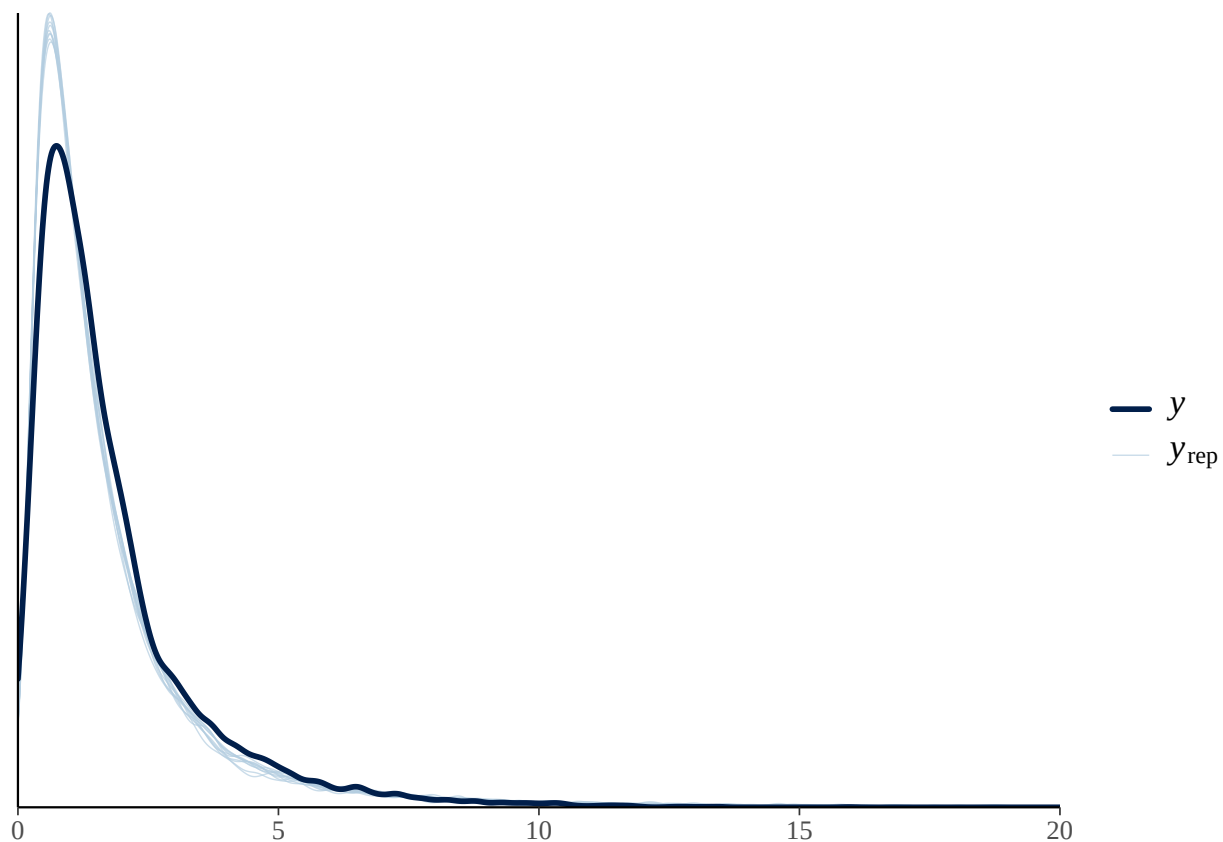
```
##
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS

```
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Per the summary above, the model displayed clean convergence, with all R-hat values either at or near 1.00. The new section, “Multilevel Hyperparameters”, quantifies the regional variation: the standard deviation of the region intercepts (0.37 on the log scale) is clearly non-zero, which confirms that regions have a genuine difference in baseline activity. The strong negative correlation between region intercepts and lag-slopes (-0.98) shows that regions with higher baseline activity tend to rely less on recent momentum, while lower-baseline regions swing more with it. Among the national-average fixed effects, the lagged elderly share remains credibly negative, consistent with the national model, while the pediatric share is no longer credibly different from zero. This suggests its apparent national signal is partially absorbed once regional structure is accounted for.

```
pp_check(fit_hier) + xlim(0, 20)
```



As shown in the posterior predictive check above, simulated data closely track the observed distribution across the full range, confirming the Gamma likelihood remains well-specified on the regional data.

Regional Variation

```
region_effects = ranef(fit_hier)$region[, , "Intercept"] %>%
  as.data.frame() %>%
  rownames_to_column("region") %>%
  rename(intercept_dev = Estimate) %>%
  select(region, intercept_dev)
region_effects
```

```
##      region intercept_dev
## 1   Region 1  -0.42908767
## 2  Region 10  -0.21941169
## 3   Region 2   0.27015223
## 4   Region 3   0.28956389
## 5   Region 4  -0.04204672
## 6   Region 5  -0.12840030
## 7   Region 6   0.51326771
## 8   Region 7  -0.28752125
## 9   Region 8  -0.36243985
## 10  Region 9   0.38907682
```

The intercept deviations shown above rank the regions by baseline flu intensity on the log scale, where positive values indicate hotter-than-average regions. Region 6 (which covers the south-central states) is the hottest, while Region 1 is the coolest. Because this ranking is estimated from the full time series rather than any individual week, it is more reliable than the snapshot a few raw observations would suggest.

```
postPredHier = posterior_predict(fit_hier, newdata = test_df)

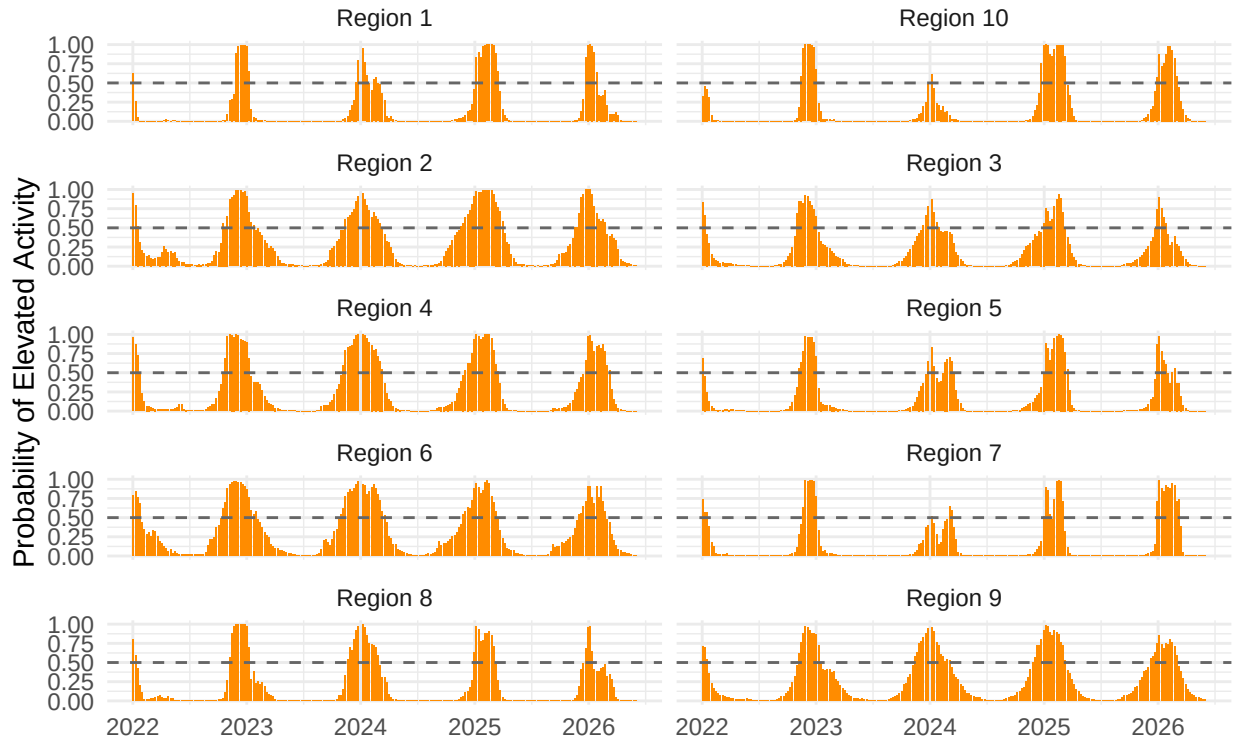
test_df = test_df %>% mutate(
  pred = colMeans(posterior_epred(fit_hier, newdata = test_df)),
  risk = colMeans(postPredHier > threshold)
)

test_df = test_df %>%
  mutate(date = as.Date(paste(year, week, 1, sep = "-"), "%Y-%U-%u"))
```

```
ggplot(test_df, aes(date, risk)) +
  geom_col(fill = "darkorange") +
  geom_hline(yintercept = 0.5, linetype = "dashed", col = "grey40") +
  facet_wrap(~ region, ncol = 2) +
  labs(title = "Estimated Weekly Risk of Elevated Flu Activity by HHS Region",
       subtitle = paste0("P(next-week wILI > ", threshold, "%), held-out seasons"),
       x = NULL, y = "Probability of Elevated Activity") +
  theme_minimal()
```


Estimated Weekly Risk of Elevated Flu Activity by HHS Region

P(next-week wILI > 3%), held-out seasons



Faceting the risk estimates by region reveals the payoff of the hierarchical model. Every region shows the expected seasonal pulse, but with clear differences in shape and intensity. Hotter regions such as 2 and 6 produce broad, sustained blocks of elevated risk across many consecutive weeks, while colder regions such as 1 and 8 show narrower, more spiky pulses. These differences, which the national model could not show, are what the regional structure was built to capture.

Does the Hierarchical Structure Improve Prediction?

The test of whether the regional structure justifies its added complexity is a leave-one-out cross-validation comparison against an otherwise identical model with no regional term, fit on the same data.

```
fit_flat = brm(
  bf(wili_next_pos ~ wili_lag1 + wili_lag2 + sinYear + cosYear +
    share_peds_lag1 + share_elderly_lag1),
  data = train_df,
  family = Gamma(link = "log"),
  chains = 4, cores = 4, iter = 2000, seed = 123,
  refresh = 0
)
```

```
loo_hier = loo(fit_hier)
loo_flat = loo(fit_flat)
loo_compare(loo_hier, loo_flat)
```

```
##           elpd_diff se_diff
## fit_hier      0.0      0.0
## fit_flat -884.8     49.9
```

The result is undeniable. The hierarchical model outperforms the flat model by approximately 885 units of expected log predictive density ($SE \approx 50$), which is a difference of around 18 standard errors. For reference, the commonly used threshold is 2 standard errors. Modeling regional structure is not a marginal refinement but a substantial improvement in predictive accuracy. With nearly 12,000 observations, large differences are detectable with high precision, so the magnitude is best interpreted as decisive rather than literal. This result confirms that influenza activity is meaningfully different across regions, a conclusion that the national model could not have reached.

Conclusion

This extension set out to determine whether modeling influenza activity hierarchically across the ten HHS regions improves on the single national model. Using the regional ILINet data and a Gamma regression with region-level intercepts and slopes, the model captured substantial variation in both baseline flu intensity and week-to-week momentum across regions, while partial pooling kept each region’s estimates stable. Leave-one-out cross-validation showed the hierarchical model to be a definite improvement over an equivalent non-hierarchical model, confirming that regional differences carry real predictive value. The resulting risk curves for each region are the practical output of this extension, as they show how the timing and intensity of elevated-activity risk differ across the country in a way the national model could not show.

Several extensions still remain. Allowing more coefficients to vary by region, incorporating the pandemic-era indicator, or modeling spatial correlation between neighboring regions could each refine this approach further. The model could also be extended to forecast multiple weeks ahead, or paired with the national model to provide forecasts at both scales simultaneously.